

Analyzing data variability in EPDs of crushed stone and asphalt

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Introduction

Demand for EPDs is relatively high nowadays in the construction sector (Konradsen, et al. 2024). The growing trend of EPD comparisons emphasizes the need for construction material stakeholders to adopt a more harmonized methodology to ensure consistent and reliable assessments across different materials (AzariJafari, et al. 2021), (Bribián, Capilla og Usón 2011). It is important to note that EPDs are developed in accordance with product category rules (PCRs) (Konradsen, et al. 2024). However, there is a risk of inconsistency and possible rule conflicts that could influence the compatibility of EPDs since different programme operators publish PCRs. Thus, using an EPD of a specific product can present significantly different impact results when following one or another guideline. It is stated that the EPDs are not highly accurate and comparable, highlighting the issue with the harmonization of the EPD system (Konradsen, et al. 2024).

The main objective of this report is to analyze various EPDs for crushed stone and asphalt by comparing their GWP (fossil emissions) for the production phase (A1-A3). The analysis focuses on how emissions are reported across various crushing stages, highlighting any variations in the way these emissions are declared for different product types and producers. By comparing these emissions, the report aims to discover potential deviations and inconsistencies in the reporting process, which are essential for making accurate, informed decisions regarding the declared emissions of these products.

Analysis of EPDs for crushed stone

In an EPD for crushed stone, different crushing stages are directly linked to the type of crushed stone product being produced, as the number of crushing stages influences both the physical properties of the final product (e.g., size and quality) and its environmental impact. The EPD ensures transparency by detailing how these stages vary for each product type, enabling stakeholders to make informed decisions about the environmental footprint of each product. The complexity of emissions declared in EPDs for crushed stone underscores the need for careful interpretation.

The assessment includes comparisons between different crushing stages across 30 EPDs of crushed stone from the Norwegian market. All the EPDs are published by EPD-Norway, a program operator for Environmental Product Declaration (EPD) type III according to ISO 14025. The EPD data were taken from:

- Viken Miljøpark AS ([NEPD-7328-6713-NO](#); [NEPD-3904-2863-NO](#))
- Ottar Skjold Maskin AS ([NEPD-7290-6689-NO](#))
- Veidekke Industri AS ([NEPD-7197-6598-NO](#); [NEPD-7196-6601-NO](#); [NEPD-6283-5545-NO](#); [NEPD-6282-5543-NO](#); [NEPD-6167-5431-NO](#); [NEPD-4543-3807-NO](#); [NEPD-4364-3593-NO](#); [NEPD-4347-3579-NO](#); [NEPD-4345-3578-NO](#); [NEPD-4346-3577-NO](#); [NEPD-4344-3576-NO](#);
- DC Eikefet Aggregates AS ([NEPD-7052-6450-NO](#))
- Skolt Pukkverk AS ([NEPD-6906-6299-NO](#); [NEPD-3856-2810-NO](#))
- Oddmund Groven AS ([NEPD-6877-6259-NO](#))
- Feiring Bruk AS ([NEPD-6392-5650-NO](#); [NEPD-6391-5651-NO](#); [NEPD-4741-3993-NO](#))
- Franzefoss Pukk AS ([NEPD-5907-5181-NO](#); [NEPD-5920-5189-NO](#); [NEPD-5189-4495-NO](#); [NEPD-5187-4497-NO](#); [NEPD-5186-4498-NO](#); [NEPD-5184-4493-NO](#);
- Bjelland AS ([NEPD-4219-3444-NO](#))
- Hamar Pukk og Gruss AS ([NEPD-3914-2873-NO](#))
- Rudshøgda Pukkverk AS ([NEPD-3905-2864-NO](#))

During the production of different types of crushed stone products, emissions are categorized into specific crushing stages, commonly referred to as Stage 0, 1, 2, 3, and, in some cases Stage 4. These crushing stages are not necessarily harmonized across all EPDs. Different EPDs may group, combine, define, and include stages differently. Therefore, when analyzing EPDs for crushed stone, it is essential to interpret how each crushing stage is reported carefully and for which products the stages are representative to ensure alignment with the intended application or comparison criteria.

Variability in declared emissions across stages involves differences in the number of crushing steps required for various product types. Products requiring additional refinement or finer granularity naturally have higher emissions. Differences in emissions may also stem from variations in product-specific requirements (e.g., size, application).

The variations in declared fossil GWP for modules A1-A3 for the included crushing stages in the 30 EPDs are presented below:

- Stage 0, which represents the initial phase of raw material extraction, has the lowest variation in emissions, with a minimum of 0.4 kg CO₂e/ton and a maximum of 3.4 kg CO₂e/ton, and an average of 1.5 kg CO₂e/ton.
- Stage 1 presents emissions, ranging from 0.7 to 4.5 kg CO₂e/ton, with an average of 2.3 kg CO₂e/ton.
- Stage 2 has the highest emissions variability, with values spanning from 1.7 to 5.9 kg CO₂e/ton and an average of 3.1 kg CO₂e/ton.
- Similarly, Stage 3 emissions range from 1.8 to 4.6 kg CO₂e/ton, with an average of 2.9 kg CO₂e/ton, suggesting a consistent energy demand for the processes in this stage, despite some differences between producers.

In addition, approximately 9 EPDs were found to have larger deviations, which need to be further investigated. For example, the EPD ([NEPD-4273-3505-NO](#)) shows lower emissions in crushing stages 3-4 compared to crushing stages 1-2, which should not be possible since emissions from previous steps shall be added to the next stage. This seems to be the current practice, even though it is not explicitly described in either the PCR or the EPDs. Furthermore, the EPDs ([NEPD-4189-3413-NO](#), [NEPD-5770-5054-NO](#)) combine two crushing stages and declare a one value. The following EPDs have added an additional stage that usually describes washing (“vaskeverk”), but also combined crushing stages ([NEPD-5188-4496-NO](#); [NEPD-5919-5190-NO](#); [NEPD-4189-3413-NO](#)). The EPDs ([NEPD-5898-5170-NO](#); [NEPD-5185-4499-NO](#) and [NEPD-4200-3429-NO](#)) include descriptions of declared stages that differ from standard practices, and certain stages are declared without explicitly referring to the corresponding stages.

The data were directly extracted from the EPDs and classified into an Excel file for analysis. Additionally, some EPDs were tested using the LCA software One Click LCA as part of a quality check. It was observed that the majority of EPDs include declared emissions separated into the different crushing stages. It is, however, hard to compare the different products/qualities because they do not necessarily require the same number of crushing stages across the producers.

Analysis of EPDs for asphalt

When comparing asphalt products with similar characteristics, it is essential to account for variations in material composition. For example:

- Group 1: 9 EPDs from Nord Vei & Asfalt AS, Brani AS and Velde Produksjon AS, with similar characteristics (containing 5–6% bitumen and 94–95% aggregate), the GWP (A1–A3) fossil emissions range from 17.9 to 83.1 kg CO₂e/ton.
- Group 2: 7 EPDs from Skanska Industrial Solutions AB for hot mix asphalt used in road construction with around 40% reclaimed asphalt included, fossil emissions are in the range from 11.3 to 21.9 kg CO₂e/ton.

These findings present the significant variability in emissions associated with asphalt production, ranging from as low as 11.3 kg CO₂e/ton to as high as 83.1 kg CO₂e/ton, depending on the material composition and production processes. The first group relies on virgin materials and conventional processes, leading to a higher CO₂e impact, while the second group includes more reclaimed asphalt, resulting in a much lower GWP.

Conclusions and recommendations

The emissions from the production of crushed stone vary significantly across the different crushing stages (Stages 0–3). Emissions reported in EPDs are not harmonized, as some EPDs combine emissions from multiple stages, exclude certain stages, or introduce additional stages for specific processes like washing (e.g., vaskeverk). Furthermore, deviations in reporting have been observed, such as cases where later stages show lower emissions than earlier ones. These inconsistencies, along with insufficient detail in some EPDs, make it challenging to compare products accurately. For both experts and non-experts, interpreting the data presented in EPDs can be challenging, particularly when emissions are expressed across multiple stages, referring to many different product types that vary significantly between different producers.

To address these issues, it is essential to standardize reporting practices by introducing clearer Product Category Rules (PCRs) for crushed stone EPDs. The PCRs should introduce the accumulation of emissions across stages, ensuring that later stages include emissions from earlier ones. This should be clearly described in the EPD. The PCRs should also require a clear separation and detailed description of each crushing stage to prevent emissions from being combined/aggregated into a single value. Additional processes, such as washing, should be transparently and clearly reported as separate stages. Finally, a harmonized categorization of product types should be defined and used across all EPDs to enable the comparison of environmental performance between products. Stricter standardized rules and requirements for performing EPDs will help to differentiate between variations due to methodological disparities and those due to real differences in performance of the products. In the EPDs for asphalt, a high potential for emission reductions is found through higher reclaimed content.

References

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